Part 2

THE PLAN!

- ❑ Causation Among Variables

CAUSATION AMONG VARIABLES

- Today, we look into
    1. Causal Relativity
    2. Causal Assignments and Response Structures
    3. Test Pairs
    4. Definition of Direct Cause
    5. Interacting Causes
    6. Response Structure Uniformity

CAUSATION AMONG VARIABLES

- (1) Causal Relativity
  - Causal generalizations are only meaningful *relative* to a given context.
    1. They are relative to a set of background conditions that are usually unstated.
    2. Claims about the causal relations between two variables X and Y are relative to the set of variables Z that are explicitly under discussion.

  - **Examples:**
    - The Homenet Study
    - The Sickle Cell Gene

CAUSATION AMONG VARIABLES

- (1) Causal Relativity
  - Claims about the direct causal relationships between one variable X and another Y must also be judged relative to the set of variables Z that are explicitly under consideration..
    - Example: Lightning Matches
      - The set of variables under consideration is called the causal system.

CAUSATION AMONG VARIABLES

- (2.1) Causal Assignments
  - Intervening on all the variables in a system besides an effect Y is to make a *causal assignment*.
  - *Causal assignment* is one particular assignment of values to all the variables in a given causal system except one variable designated as the effect.
    - Example: The Garage Light

Assignment	Switch 1	Switch 2
1	down	down
2	down	up
3	up	down
4	up	up

- Since there are two switches, and each has two positions, there are four possible causal assignments in the causal system involving the garage light and the two light switches.
- Although the system involves three variables, we designated the garage light as the effect. We call the other two potential causes.

CAUSATION AMONG VARIABLES

- (2.1.) Causal Assignments
  - Causal assignment involves the idea of *forcing* the variables to have a particular set of values.
  - We are ***assigning values, not observing*** them.
    - Example: Malaria

    - *Please note that causal assignments are 'Interventions' not 'Observations'!!!!*

CAUSATION AMONG VARIABLES

- (2.1.) Causal Assignments
  - The number of different causal assignments for a given effect grows with both the number of potential causes in the system and the number of values for each.
    - In general, if there are N variables, then the number of causal assignments is equal to:

(# of values of Var. 1) X (# of values of Var. 2) X … X (# of values for Var. N)

Assignment	Switch 1	Switch 2
1	down	down
2	down	up
3	up	down
4	up	up

If there would be three switches, the number would be 8, with four switches the number would 16, and so on.

If the two switches had three positions each, there would be 9 assignments.

CAUSATION AMONG VARIABLES

- (2.2.) Response Structures
  - The route to a general account of causation among variables goes through tables which include all the possible causal assignments for some effect, as well as some description of the effect in each such assignment.
  - We call such tables *response structures*

Table 4: Empty Response Structure for the Garage Light

Assignment	Switch 1	Switch 2	Effect: Garage Light
1	down	down	???
2	down	up	???
3	up	down	???
4	up	up	???

Table 5: Response Structure for the Garage Light

Assignment	Switch 1	Switch 2	Effect: Garage Light
1	down	down	Off
2	down	up	On
3	up	down	Off
4	up	up	On

CAUSATION AMONG VARIABLES

- (2.2.) Response Structures
  - A response structure must contain *all* possible assignments.
  - Method for building a response structure:

1. Calculate the number of rows in the table. If there are N variables, then the number of causal assignments is equal to: (# of values of Var. 1) X (# of values of Var. 2) X … X (# of values for Var. N)
2. Begin with the rightmost potential cause column, and alternate as quickly as possible the values for the variable in the column.
3. Go one column on the left. Fill in the first value of this variable repeatedly, for one full cycle of values of the variable just to its right. Fill in the next value repeatedly, for one cycle of values of the variable to the right, etc.
4. Repeat step 3 until there are no more variables.

CAUSATION AMONG VARIABLES

- (3) Test Pairs
  - Generalization about direct causation among variables have the following form:
    - Variable X is a direct cause of variable Y relative to the set of variable Z under the conditions B.

- **Examples:** Chicken Pox; Cigarette Prices

CAUSATION AMONG VARIABLES

- (3) Test Pairs

Assignment	Switch 1	Switch 2	Effect: Garage Light
1	down	down	Off
2	down	up	On
3	up	down	Off
4	up	up	On

If there is a difference in the effect, then we have a direct causation.

- Test pair

Test Pair of Causal Assignments for X: Two causal assignments C1 and C2 are a *test pair of causal assignments* for X if and only if they are identical except for the values assigned to the variable X.

Causal assignments 1 and 2 are the same *except* for the value of Switch 2 assigned. Everything else in the system is assigned the same value (Switch 1 = down), but Switch 2 is down in assignment 1 and up in assignment 2.

The same goes for causal assignment 3 and 4, except that in this pair Switch 1 = up. Switch 2 is down in assignment 3 and up in assignment 4.

The idea is to hold everything else in the system constant, and *only* vary the potential cause.

CAUSATION AMONG VARIABLES

- (4) Definition of Direct Cause
  - Direct Cause: In a system of variables S, X is a direct cause of Y relative to S if and only there is at least one test pair of causal assignments for X in the response structure for Y across which Y differs.
    - Example: The Switch, Battery, and Light Bulb

CAUSATION AMONG VARIABLES

- (5) Interacting Causes
  - When the *influence* of one direct cause depends upon the state of another direct cause, then we say the causes ***interact***.
    - **Examples:** Train Arrivals; the Battery, Switch, and Light Bulb
  - Interacting Causes: When the influence of one direct cause on the effect depends upon the ***value*** of another direct cause, then we say that the causes ***interact***

CAUSATION AMONG VARIABLES

- Causation in Populations
  - Individuals may differ in both causal assignments and response structures. To generalize causal claims, we often assume Response Structure Uniformity: all individuals in a population share the same response structure.
  - Violations exist (e.g., beard growth differs by gender).
  - Sub-populations can be defined where uniformity holds.

Reading Exercise – No 14 (Module 2 - Causation among variables)

This module develops a framework for understanding direct causation among variables, focusing on discrete systems. It introduces key terminology and methods for identifying when one variable directly causes another.

- Causal Relativity - Causation is always contextual. Background conditions shape causal claims (e.g., diet effects differ by culture/time).
- Causal system: set of variables explicitly under discussion. A variable may be a direct cause in one system but only an indirect cause in another.
- Causal reasoning takes place relative to background conditions as well as a specified set of variables (called the causal system).
- Direct Cause - in a system of variables S, X is a ***direct cause*** of Y relative to S if and only if there is at least one *test pair* of *causal assignments for X* in the *response structure for Y* across which Y differs.
- Causal assignment - one way to reliably isolate which, among many potential causes of Y, is a real cause is to intervene on all the variables in a system besides Y and give them every possible value. Intervening on all the variables in a system besides an effect Y is to make a ***causal assignment***. *Please note that causal assignments are 'Interventions' not 'Observations'.*
- Response Structure - route to a general account of causation among variables goes through tables which include all the possible causal assignments for some effect, as well as some description of the effect in each such assignment. We call such tables ***response structures***.
- If there are any two causal assignments that are identical except for the value assigned to X, and there is a difference in the effect Y, ***then X is a direct cause of Y.***
  - **Test Pair of Causal Assignments for X** - Two causal assignments C1 and C2 are a test pair of causal assignments for X if and only if they are identical except for the values assigned to variable X.
  - **Direct Cause** - In a system of variables S, X is a direct cause of Y relative to S if and only if there is at least one test pair of causal assignments for X in the response structure for Y across which Y differs.
- **Causation in Populations -** Individuals may differ in both **causal assignments** and **response structures**. To generalize causal claims, we often assume **Response Structure Uniformity**: all individuals in a population share the same response structure.
  - Violations exist (e.g., beard growth differs by gender).
  - Sub-populations can be defined where uniformity holds.

Reading Exercise – No 14 (Causation among variables)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

A - Reflect and discuss - Causal Relativity: Background Conditions

Question No 01 – Consider the claim: “Drinking at least 3 glasses of tap water a day will improve your health.” **Which of the following are background conditions that this claim must be judged against (choose one):**

A. Whether your community has enough water

B. Whether your community’s water supply contains toxic chemicals.

C. Whether your community is religious or not.

D. None of the above.

Reading Exercise – No 14 (Causation among variables)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

A - Reflect and discuss - Causal Relativity: Background Conditions

Question No 01 – Consider the claim: "Drinking at least 3 glasses of tap water a day will improve your health." **Which of the following are background conditions that this claim must be judged against (choose one):**

A. Whether your community has enough water

B. Whether your community's water supply contains toxic chemicals.

C. Whether your community is religious or not.

D. None of the above.

Answer B. The level of toxic chemicals in the tap water is a relevant background condition for this causal claim. If the tap water has a high level of toxic chemicals, drinking 3 glasses a day will almost surely damage your health. If not, it will probably improve your health.

Reading Exercise – No 14 (Causation among variables)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

B - Reflect and discuss - Causal Relativity: The Set of Variables

Question No 02 – In which of the causal systems below is Exposure a direct cause of Rash? (You may select more than one answer.)

A. Exposure to Chicken Pox [yes, no], High Fever [yes, no], Chicken Pox Rash [yes, no].

B. Exposure to Chicken Pox [yes, no], Infection with Chicken Pox [yes, no], Chicken Pox Rash [yes, no].

C. Exposure to Chicken Pox [yes, no], Chicken Pox Rash [yes, no].

D. None of the above

Reading Exercise – No 14 (Causation among variables)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

B - Reflect and discuss - Causal Relativity: The Set of Variables

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A. **Exposure to Chicken Pox [yes, no], High Fever [yes, no], Chicken Pox Rash [yes, no].**

B. Exposure to Chicken Pox [yes, no], Infection with Chicken Pox [yes, no], Chicken Pox Rash [yes, no].

C. **Exposure to Chicken Pox [yes, no], Chicken Pox Rash [yes, no].**

D. None of the above

Answer - A and C.

In the system with only two variables: Exposure and Rash, Exposure is a direct cause of Rash. Also, Exposure is a direct cause of Rash in the system involving Exposure, High Fever, and Rash. We are assuming that it is not the high fever which causes the rash, (when you have the flu you get high fever but not chicken pox rash), so exposure must still directly cause the rash relative to that system.

Reading Exercise – No 14 (Causation among variables)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

B - Reflect and discuss - Causal Relativity: The Set of Variables

Question No 03 – When we specify a causal system, we must include (choose one):

A. All of the potential causes we know about.

B. All of the potential causes.

C. All of the actual causes.

D. None of the above.

Reading Exercise – No 14 (Causation among variables)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

B - Reflect and discuss - Causal Relativity: The Set of Variables

Question No 03 – When we specify a causal system, we must include (choose one):

A. All of the potential causes we know about.

B. All of the potential causes.

C. All of the actual causes.

D. None of the above.

Answer - D.

A causal system is just any set of variables. The relations that obtain these variables depends on the set specified, but any set can be specified.

Reading Exercise – No 14 (Causation among variables)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

C - Reflect and discuss - Causal assignments

Question No 04 – In a causal assignment (choose one):

A. The values of each of the variables in the causal system besides the effect are observed

B. The values of each of the variables in the causal system besides the effect are assigned by an intervention

C. Every conceivable cause of the effect is included

D. The values of each of the variables in the causal system, including the effect, are assigned by an intervention

Reading Exercise – No 14 (Causation among variables)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

C - Reflect and discuss - Causal assignments

Question No 04 – In a causal assignment (choose one):

A. The values of each of the variables in the causal system besides the effect are observed

B. The values of each of the variables in the causal system besides the effect are assigned by an intervention

C. Every conceivable cause of the effect is included

D. The values of each of the variables in the causal system, including the effect, are assigned by an intervention

The correct answer is B.

In a causal assignment the values of the variables are assigned by an intervention rather than merely observed.

Reading Exercise – No 14 (Causation among variables)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

C - Reflect and discuss - Causal assignments

Question No 05 – Suppose that our causal system consists of the variables Gas Tank Full [yes, no], Key Turned [yes, no], and Engine Starts [yes, no], where Engine Starts is the effect. **Which of the following are causal assignments? (choose one)**

A. Cutting the cables from the battery to the spark plugs.

B. Putting a charged battery and a full tank of gas in the car.

C. Turning the key.

D. Emptying the gas tank and turning the key.

Reading Exercise – No 14 (Causation among variables)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

C - Reflect and discuss - Causal assignments

Question No 05 – Suppose that our causal system consists of the variables Gas Tank Full [yes, no], Key Turned [yes, no], and Engine Starts [yes, no], where Engine Starts is the effect. **Which of the following are causal assignments? (choose one)**

A. Cutting the cables from the battery to the spark plugs.
B. Putting a charged battery and a full tank of gas in the car.
C. Turning the key.
D. Emptying the gas tank and turning the key.

The correct answer is D.

A causal assignment does not assign a value to the effect.

Reading Exercise – No 14 (Causation among variables)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

D - Reflect and discuss - Causal Assignments: Intervention vs. Observation (Causal Assignments are Interventions, not Observations)

Question No 06 – Specifying a causal assignment requires specifying (choose one):

A. The background conditions.

B. A variable designated as the cause

C. An assignment of a value to each variable in the system besides the effect

D. All of the above

E. None of the above

Reading Exercise – No 14 (Causation among variables)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

D - Reflect and discuss - Causal Assignments: Intervention vs. Observation (Causal Assignments are Interventions, not Observations)

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A. The background conditions.

B. A variable designated as the cause

C. An assignment of a value to each variable in the system besides the effect

D. All of the above

E. None of the above

Reading Exercise – No 14 (Causation among variables)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

D - Reflect and discuss - Causal Assignments: Intervention vs. Observation (Causal Assignments are Interventions, not Observations)

Question No 07 – Assume the causal system: {Bitten by Mosquito, Had Inoculation, Has Sickle Cell Gene, Drinks, Gets malaria}. **Which of the following is a causal assignment for the effect malaria? (choose one)**

A. Sam was bitten by a mosquito, Sam did not have a malaria inoculation, Sam lives in America.

B. Sam was bitten by a mosquito, Sam did have a malaria inoculation, Sam has the sickle cell gene.

C. Sam was bitten by a mosquito, Sam did have a malaria inoculation, Sam has the sickle cell gene, and Sam gets Malaria.

D. Sam was bitten by a mosquito, Sam did have a malaria inoculation, Sam has the sickle cell gene, and Sam drinks Gin and Tonics.

Reading Exercise – No 14 (Causation among variables)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

D - Reflect and discuss - Causal Assignments: Intervention vs. Observation (Causal Assignments are Interventions, not Observations)

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B. Sam was bitten by a mosquito, Sam did have a malaria inoculation, Sam has the sickle cell gene.

C. Sam was bitten by a mosquito, Sam did have a malaria inoculation, Sam has the sickle cell gene, and Sam gets Malaria.

D. Sam was bitten by a mosquito, Sam did have a malaria inoculation, Sam has the sickle cell gene, and Sam drinks Gin and Tonics.

Reading Exercise – No 14 (Causation among variables)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

D - Reflect and discuss - Causal Assignments: Intervention vs. Observation (Causal Assignments are Interventions, not Observations)

Question No 08 – Consider this set of variables (and values): Smokes [Yes, No]; Nicotine-Stained Fingers [Yes, No]; and Lung Cancer [Yes, No]. **If we designate Lung Cancer as our effect variable, then which of following is a legitimate causal assignment? (choose one)**

A. Smokes = yes, Lung Cancer = No

B. Smokes = yes, Nicotine-Stained Fingers = Yes, Lung Cancer = No

C. Smokes = yes

D. Nicotine-Stained Fingers = No, Lung Cancer = No

E. Smokes = yes, Nicotine-Stained Fingers = No

Reading Exercise – No 14 (Causation among variables)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

D - Reflect and discuss - Causal Assignments: Intervention vs. Observation (Causal Assignments are Interventions, not Observations)

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A. Smokes = yes, Lung Cancer = No

B. Smokes = yes, Nicotine-Stained Fingers = Yes, Lung Cancer = No

C. Smokes = yes

D. Nicotine-Stained Fingers = No, Lung Cancer = No

E. Smokes = yes, Nicotine-Stained Fingers = No

Reading Exercise – No 14 (Causation among variables)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

E - Reflect and discuss – The number of Causal assignments

Question No 09 – Consider a causal system with two variables:

- TV [watch none, watch some, watch a lot]
- Fight [never, some, a lot]

If we consider the response structure in which the variable Fight is the effect, how many causal assignments are there? (choose one)

A. 2

B. 3

C. 4

D. 9

Reading Exercise – No 14 (Causation among variables)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

E - Reflect and discuss – The number of Causal assignments

Question No 09 – Consider a causal system with two variables:

- TV [watch none, watch some, watch a lot]
- Fight [never, some, a lot]

If we consider the response structure in which the variable Fight is the effect, how many causal assignments are there? (choose one)

A. 2

B. 3

C. 4

D. 9

Answer - B.

The response structure for Fight would have three possible causal assignments:

- **TV = watch none,**
- **TV = watch some, and**
- **TV = watch a lot.**

Reading Exercise – No 14 (Causation among variables)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

E - Reflect and discuss – The number of Causal assignments

Question No 10 – Consider a causal system with three variable.

- Meals eaten per day [1, 2, 3]
- Workouts per day [1, 2]
- Weight [Loss, Gain]

Consider the response structure in which the variable Weight is the effect. **How many causal assignments are there? (choose one)**

A. 2

B. 3

C. 4

D. 6

Reading Exercise – No 14 (Causation among variables)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

E - Reflect and discuss – The number of Causal assignments

Question No 10 – Consider a causal system with three variable.

- Meals eaten per day [1,2,3]
- Workouts per day [1,2]
- Weight [Loss, Gain]

Consider the response structure in which the variable Weight is the effect. **How many causal assignments are there? (choose one)**

A. 2

B. 3

C. 4

D. 6

Answer - D. There are six possible causal assignments:

Causal Assignment	Meals Eaten	Workouts
1	1	1
2	1	2
3	2	1
4	2	2
5	3	1
6	3	2

Reading Exercise – No 14 (Causation among variables)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

F - Reflect and discuss – Defining Direct Causation Among Variable – Test Pairs 1

Question No 11 –

Assignment	Switch 1	Switch 2	Effect: Garage Light
1	down	down	off
2	down	up	on
3	up	down	off
4	up	up	on

According to this table, which pairs of causal assignments are the same except for the value that the variable Switch 2 is assigned? (Check all that apply)

A. 1 and 2.

B. 1 and 3.

C. 1 and 4.

D. 2 and 3.

E. 2 and 4.

F. 3 and 4.

Reading Exercise – No 14 (Causation among variables)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

F - Reflect and discuss – Defining Direct Causation Among Variable – Test Pairs 1

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1	down	down	off
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4	up	up	on

According to this table, which pairs of causal assignments are the same except for the value that the variable Switch 2 is assigned? (Check all that apply)

A. 1 and 2.

B. 1 and 3.

C. 1 and 4.

D. 2 and 3.

E. 2 and 4.

F. 3 and 4.

Answer - A and F.

The pair 1-2 is correct, and so is the pair 3-4, and no other pairs are the same except for Switch 2.

Reading Exercise – No 14 (Causation among variables)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

F - Reflect and discuss – Defining Direct Causation Among Variable – Test Pairs 2

Question No 12 – Consider this response structure:

Causal Assignment	Physical Therapy	Medication	Recovers
1	Yes	Yes	Yes
2	Yes	No	No
3	No	Yes	No
4	No	No	No

Which of the following are test pairs for Medication? (choose one)

A. 3 and 4

B. 1 and 3

C. 2 and 4

D. 2 and 3

Reading Exercise – No 14 (Causation among variables)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

F - Reflect and discuss – Defining Direct Causation Among Variable – Test Pairs 2

Question No 12 – Consider this response structure:

Causal Assignment	Physical Therapy	Medication	Recovers
1	Yes	Yes	Yes
2	Yes	No	No
3	No	Yes	No
4	No	No	No

Which of the following are test pairs for Medication? (choose one)

A. **3 and 4**

B. 1 and 3

C. 2 and 4

D. 2 and 3

Reading Exercise – No 14 (Causation among variables)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

F - Reflect and discuss – Defining Direct Causation Among Variable – Test Pairs 2

Question No 13 – Consider the following response structure for Passed Exam:

Causal Assignment	Slept 8 Hours	Studied	Passed Exam
1	Yes	Yes	No
2	Yes	No	No
3	No	Yes	Yes
4	No	No	No

Which of the following are test pairs for Slept 8 Hours? (choose one)

A. 1 and 2

B. 2 and 3

C. 1 and 3

D. 3 and 4

Reading Exercise – No 14 (Causation among variables)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

F - Reflect and discuss – Defining Direct Causation Among Variable – Test Pairs 2

Question No 13 – Consider the following response structure for Passed Exam:

Causal Assignment	Slept 8 Hours	Studied	Passed Exam
1	Yes	Yes	No
2	Yes	No	No
3	No	Yes	Yes
4	No	No	No

Which of the following are test pairs for Slept 8 Hours? (choose one)

A. 1 and 2

B. 2 and 3

C. **1 and 3**

D. 3 and 4

The correct answer is C.

Assignments 1 and 2 don't differ on Slept 8 Hours.

Reading Exercise – No 14 (Causation among variables)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

G - Reflect and discuss – Defining Direct Causation Among Variables 1

Question No 14 –

Assignment	Switch 1	Switch 2	Effect: Garage Light
1	down	down	off
2	down	up	on
3	up	down	off
4	up	up	on

Referring to the response structure above, which pairs of causal assignments are test pairs for Switch 1. (Check all that apply.)

A. 1 and 2.

B. 1 and 3.

C. 1 and 4.

D. 2 and 3.

E. 2 and 4.

F. 3 and 4.

Reading Exercise – No 14 (Causation among variables)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

G - Reflect and discuss – Defining Direct Causation Among Variables 1

Question No 14 –

Assignment	Switch 1	Switch 2	Effect: Garage Light
1	down	down	off
2	down	up	on
3	up	down	off
4	up	up	on

Referring to the response structure above, which pairs of causal assignments are test pairs for Switch 1. (Check all that apply.)

A. 1 and 2.

B. 1 and 3.

C. 1 and 4.

D. 2 and 3.

E. 2 and 4.

F. 3 and 4.

Answer - B and E.

The pairs 1 - 3 and 2 - 4 are the only test pairs for Switch 1.

Reading Exercise – No 14 (Causation among variables)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

G - Reflect and discuss – Defining Direct Causation Among Variables 1

Question No 15 –

Assignment	Switch 1	Switch 2	Effect: Garage Light
1	down	down	off
2	down	up	on
3	up	down	off
4	up	up	on

Across which test pairs of causal assignments for Switch 1 is the garage light (the effect) different? (choose all that apply.)

A. Causal assignments 1 and 3.

B. Causal assignments 2 and 4.

C. None of the above

Reading Exercise – No 14 (Causation among variables)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

G - Reflect and discuss – Defining Direct Causation Among Variables 1

Question No 15 –

Assignment	Switch 1	Switch 2	Effect: Garage Light
1	down	down	off
2	down	up	on
3	up	down	off
4	up	up	on

Across which test pairs of causal assignments for Switch 1 is the garage light (the effect) different? (choose all that apply.)

A. Causal assignments 1 and 3.

B. Causal assignments 2 and 4.

C. None of the above

Answer - C.

The garage light is off in both assignments 1 and 3, and the garage light is on in both assignments 2 and 4

Reading Exercise – No 14 (Causation among variables)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

I - Reflect and discuss – Variable Causation Quiz

Question No 16 – X is a direct cause of Y if and only if (choose one):

A. Y differs in every test pair for X

B. There is at least one test pair for X in the response structure for Y across which Y differs

C. There is at least one test pair for X in the response structure for Y across which Y does not differ

D. There is at least one test pair for X in the response structure for Y

Reading Exercise – No 14 (Causation among variables)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

I - Reflect and discuss – Variable Causation Quiz

Question No 16 – X is a direct cause of Y if and only if (choose one):

A. Y differs in every test pair for X

B. There is at least one test pair for X in the response structure for Y across which Y differs

C. There is at least one test pair for X in the response structure for Y across which Y does not differ

D. There is at least one test pair for X in the response structure for Y

Reading Exercise – No 14 (Causation among variables)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

I - Reflect and discuss – Variable Causation Quiz

Question No 17 – Consider the following response structure:

Causal Assignment	Starter Broken	Battery Charged	Car Starts
1	Yes	Yes	No
2	Yes	No	No
3	No	Yes	Yes
4	No	No	No

Starter Broken is a direct cause of Car Starts in this response structure because (choose one):

A. 2 and 3 is a test pair for Starter Broken in which Car Starts differs

B. 3 and 4 is a test pair for Starter Broken in which Car Starts differs

C. 1 and 3 is a test pair for Starter Broken in which Car Starts differs

D. 2 and 4 is a test pair for Starter Broken in which Car Starts differs

Reading Exercise – No 14 (Causation among variables)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

I - Reflect and discuss – Variable Causation Quiz

Question No 17 – Consider the following response structure:

Causal Assignment	Starter Broken	Battery Charged	Car Starts
1	Yes	Yes	No
2	Yes	No	No
3	No	Yes	Yes
4	No	No	No

Starter Broken is a direct cause of Car Starts in this response structure because (choose one):

A. 2 and 3 is a test pair for Starter Broken in which Car Starts differs

B. 3 and 4 is a test pair for Starter Broken in which Car Starts differs

C. 1 and 3 is a test pair for Starter Broken in which Car Starts differs

D. 2 and 4 is a test pair for Starter Broken in which Car Starts differs

The correct answer is C.

(B. 3 and 4 is not a test pair for Starter Broken, since the value of Starter Broken is the same.)

Reading Exercise – No 14 (Causation among variables)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

I - Reflect and discuss – Variable Causation Quiz

Question No 18 – Consider the following response structure:

Causal Assignment	X	Y	Z
1	Yes	Yes	Yes
2	Yes	No	Yes
3	No	Yes	No
4	No	No	No

In this response structure, which variables are direct causes of Z? (choose one)

A. X only

B. Y only

C. Both X and Y

D. Neither X nor Y

Reading Exercise – No 14 (Causation among variables)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

I - Reflect and discuss – Variable Causation Quiz

Question No 18 – Consider the following response structure:

Causal Assignment	X	Y	Z
1	Yes	Yes	Yes
2	Yes	No	Yes
3	No	Yes	No
4	No	No	No

In this response structure, which variables are direct causes of Z? (choose one)

A. X only
B. Y only
C. Both X and Y
D. Neither X nor Y

Answer - A.

1 and 3 is a test pair for X in which the value of Z differs.

Reading Exercise – No 14 (Causation among variables)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

J - Reflect and discuss – Causation: Subtleties 1

Question No 19 –

Response Structure for the Light Bulb

Assignment	Switch	Battery	Effect: Light Bulb
1	down	charged	Off
2	down	uncharged	Off
3	up	charged	On
4	up	uncharged	Off

Referring to table titled: Response Structure for the Light Bulb, which pairs of causal assignments are test pairs for the Switch? (choose all that apply.)

A. 1 and 2.

B. 1 and 3.

C. 1 and 4.

D. 2 and 3.

E. 2 and 4.

F. 3 and 4.

Reading Exercise – No 14 (Causation among variables)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

J - Reflect and discuss – Causation: Subtleties 1

Question No 19 –

Response Structure for the Light Bulb

Assignment	Switch	Battery	Effect: Light Bulb
1	down	charged	Off
2	down	uncharged	Off
3	up	charged	On
4	up	uncharged	Off

Referring to table titled: Response Structure for the Light Bulb, which pairs of causal assignments are test pairs for the Switch? (choose all that apply.)

A. 1 and 2.

B. 1 and 3.

C. 1 and 4.

D. 2 and 3.

E. 2 and 4.

F. 3 and 4.

Reading Exercise – No 14 (Causation among variables)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

K - Reflect and discuss – Causation: Subtleties 2

Question No 20 – Referring to the table in the reading entitled: Response Structure For The Malaria Case, **which pairs of causal assignments are test pairs for the variable Drinker? (choose all that apply.)**

A. 1 and 2.

B. 1 and 3.

C. 1 and 4.

D. 9 and 10.

E. 15 and 16.

F. 1 and 9.

Table 7: Response Structure For The Malaria Case

Assignment	Variable 1: Bitten	Variable 2: Inoculated	Variable 3: Has Gene	Variable 4: Drinker	Effect: Malaria
1	True	True	True	True	False
2	True	True	True	False	False
3	True	True	False	True	False
4	True	True	False	False	False
5	True	False	True	True	False
6	True	False	True	False	False
7	True	False	False	True	True
8	True	False	False	False	True
9	False	True	True	True	False
10	False	True	True	False	False
11	False	True	False	True	False
12	False	True	False	False	False
13	False	False	True	True	False
14	False	False	True	False	False
15	False	False	False	True	False
16	False	False	False	False	False

Reading Exercise – No 14 (Causation among variables)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

K - Reflect and discuss – Causation: Subtleties 2

Question No 20 – Referring to the table in the reading entitled: Response Structure For The Malaria Case, **which pairs of causal assignments are test pairs for the variable Drinker? (choose all that apply.)**

- **A. 1 and 2.**
- B. 1 and 3.
- C. 1 and 4.
- **D. 9 and 10.**
- **E. 15 and 16.**
- F. 1 and 9.

Table 7: Response Structure For The Malaria Case

Assignment	Variable 1: Bitten	Variable 2: Inoculated	Variable 3: Has Gene	Variable 4: Drinker	Effect: Malaria
1	True	True	True	True	False
2	True	True	True	False	False
3	True	True	False	True	False
4	True	True	False	False	False
5	True	False	True	True	False
6	True	False	True	False	False
7	True	False	False	True	True
8	True	False	False	False	True
9	False	True	True	True	False
10	False	True	True	False	False
11	False	True	False	True	False
12	False	True	False	False	False
13	False	False	True	True	False
14	False	False	True	False	False
15	False	False	False	True	False
16	False	False	False	False	False

Reading Exercise – No 14 (Causation among variables)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

K - Reflect and discuss – Causation: Subtleties 2

Question No 21 – All the following are test pairs of causal assignments for the variable Drinker. **In which pairs does the effect differ? (choose one).**

A. 1 and 2
B. 3 and 4
C. 5 and 6
D. 7 and 8
E. 9 and 10
F. 11 and 12
G. 13 and 14
H. 15 and 16
I. None of the above.

Table 7: Response Structure For The Malaria Case

Assignment	Variable 1: Bitten	Variable 2: Inoculated	Variable 3: Has Gene	Variable 4: Drinker	Effect: Malaria
1	True	True	True	True	False
2	True	True	True	False	False
3	True	True	False	True	False
4	True	True	False	False	False
5	True	False	True	True	False
6	True	False	True	False	False
7	True	False	False	True	True
8	True	False	False	False	True
9	False	True	True	True	False
10	False	True	True	False	False
11	False	True	False	True	False
12	False	True	False	False	False
13	False	False	True	True	False
14	False	False	True	False	False
15	False	False	False	True	False
16	False	False	False	False	False

Reading Exercise – No 14 (Causation among variables)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

K - Reflect and discuss – Causation: Subtleties 2

Question No 21 – All the following are test pairs of causal assignments for the variable Drinker. **In which pairs does the effect differ? (choose one).**

A. 1 and 2
B. 3 and 4
C. 5 and 6
D. 7 and 8
E. 9 and 10
F. 11 and 12
G. 13 and 14
H. 15 and 16
I. None of the above.

Table 7: Response Structure For The Malaria Case

Assignment	Variable 1: Bitten	Variable 2: Inoculated	Variable 3: Has Gene	Variable 4: Drinker	Effect: Malaria
1	True	True	True	True	False
2	True	True	True	False	False
3	True	True	False	True	False
4	True	True	False	False	False
5	True	False	True	True	False
6	True	False	True	False	False
7	True	False	False	True	True
8	True	False	False	False	True
9	False	True	True	True	False
10	False	True	True	False	False
11	False	True	False	True	False
12	False	True	False	False	False
13	False	False	True	True	False
14	False	False	True	False	False
15	False	False	False	True	False
16	False	False	False	False	False

Reading Exercise – No 14 (Causation among variables)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

K - Reflect and discuss – Causation: Subtleties 2

Question No 22 – Is the variable Drinker a direct cause of malaria, according to the response structure in table below and our definition of direct cause? (choose one)

A. Yes

B. No

C. Impossible to tell.

Table 7: Response Structure For The Malaria Case

Assignment	Variable 1: Bitten	Variable 2: Inoculated	Variable 3: Has Gene	Variable 4: Drinker	Effect: Malaria
1	True	True	True	True	False
2	True	True	True	False	False
3	True	True	False	True	False
4	True	True	False	False	False
5	True	False	True	True	False
6	True	False	True	False	False
7	True	False	False	True	True
8	True	False	False	False	True
9	False	True	True	True	False
10	False	True	True	False	False
11	False	True	False	True	False
12	False	True	False	False	False
13	False	False	True	True	False
14	False	False	True	False	False
15	False	False	False	True	False
16	False	False	False	False	False

Reading Exercise – No 14 (Causation among variables)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

K - Reflect and discuss – Causation: Subtleties 2

Question No 22 – Is the variable Drinker a direct cause of malaria, according to the response structure in table below and our definition of direct cause? (choose one)

A. Yes

B. No

C. Impossible to tell.

Table 7: Response Structure For The Malaria Case

Assignment	Variable 1: Bitten	Variable 2: Inoculated	Variable 3: Has Gene	Variable 4: Drinker	Effect: Malaria
1	True	True	True	True	False
2	True	True	True	False	False
3	True	True	False	True	False
4	True	True	False	False	False
5	True	False	True	True	False
6	True	False	True	False	False
7	True	False	False	True	True
8	True	False	False	False	True
9	False	True	True	True	False
10	False	True	True	False	False
11	False	True	False	True	False
12	False	True	False	False	False
13	False	False	True	True	False
14	False	False	True	False	False
15	False	False	False	True	False
16	False	False	False	False	False

Answer - B.

Drinker is NOT a cause of malaria, because there are NO test pairs for Drinker across which malaria is different.

Reading Exercise – No 14 (Causation among variables)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

L - Reflect and discuss – Interactions 1

Question No 23 – In the Garage Light scenario, do Switch 1 and Switch 2 interact to cause the garage light to come on? (choose one)

A. Yes

B. No

C. Not enough information to tell

Assignment	Switch 1	Switch 2	Effect: Garage Light
1	down	down	Off
2	down	up	On
3	up	down	Off
4	up	up	On

Reading Exercise – No 14 (Causation among variables)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

L - Reflect and discuss – Interactions 1

Question No 23 – In the Garage Light scenario, do Switch 1 and Switch 2 interact to cause the garage light to come on? (choose one)

A. Yes

B. No

C. Not enough information to tell

Assignment	Switch 1	Switch 2	Effect: Garage Light
1	down	down	Off
2	down	up	On
3	up	down	Off
4	up	up	On

Answer - B.

Switch 2 causes the light to come on when it is in the "on" position. Its influence does not depend on the position of Switch 1.

Reading Exercise – No 14 (Causation among variables)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

L - Reflect and discuss – Interactions 1

Question No 24 –

Causal Assignment	Starter Broken	Battery Charged	Car Starts
1	Yes	Yes	No
2	Yes	No	No
3	No	Yes	Yes
4	No	No	No

In this response structure, the variables Starter Broken [yes, no] and Battery Charged [yes, no] interactively cause Car Starts [yes, no] because (choose one):

A. Both are causes of Car Starts

B. The influence of Battery Charged on the Car Starts depends on the value of Starter Broken

C. Starter Broken is not a cause of Car Starts

D. They have the same test pairs

Reading Exercise – No 14 (Causation among variables)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

L - Reflect and discuss – Interactions 1

Question No 24 –

Causal Assignment	Starter Broken	Battery Charged	Car Starts
1	Yes	Yes	No
2	Yes	No	No
3	No	Yes	Yes
4	No	No	No

In this response structure, the variables Starter Broken [yes, no] and Battery Charged [yes, no] interactively cause Car Starts [yes, no] because (choose one):

A. Both are causes of Car Starts
B. **The influence of Battery Charged on the Car Starts depends on the value of Starter Broken**
C. Starter Broken is not a cause of Car Starts
D. They have the same test pairs

The correct answer is B.

It is possible for two variables to be causes of a third without interactively causing it.

Reading Exercise – No 14 (Causation among variables)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

M - Reflect and discuss – Response Structure Uniformity

Question No 25 – Is it likely that the population of all 1993 Ford Escorts exhibit response structure uniformity with respect to whether their gas tanks explode in rear-end collisions or not (because of a structural defect in the way that 1993 Escorts were built)? (choose one)

A. Yes

B. No

Reading Exercise – No 14 (Causation among variables)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

M - Reflect and discuss – Response Structure Uniformity

Question No 25 – Is it likely that the population of all 1993 Ford Escorts exhibit response structure uniformity with respect to whether their gas tanks explode in rear-end collisions or not (because of a structural defect in the way that 1993 Escorts were built)? (choose one)

A. **Yes**

B. No

Answer - A.

Assuming that they were all built in the same way, you can expect to see the same results for each possible causal assignment.

Reading Exercise – No 14 (Causation among variables)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

M - Reflect and discuss – Response Structure Uniformity

Question No 26 – Is it likely that the population of all 1993 Ford Escorts exhibit response structure uniformity with respect to exposure to the sun causing their upholstery to crack? (choose one)

A. Yes

B. No

Reading Exercise – No 14 (Causation among variables)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

M - Reflect and discuss – Response Structure Uniformity

Question No 26 – Is it likely that the population of all 1993 Ford Escorts exhibit response structure uniformity with respect to exposure to the sun causing their upholstery to crack? (choose one)

A. Yes

B. No

Answer - B.

Like all cars, their upholstery comes in different materials: leather, vinyl, and cloth, and these materials deteriorate and crack at different rates. This means that they have different response structures with respect to the sun causing their upholstery to crack.

Reading Exercise – No 14 (Causation among variables)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

N - Reflect and discuss – Link it to your research proposal

Many of the ideas you have learned in this handout will be relevant to your Semester Research Project. Discuss the connection of these ideas to your project here in as much detail as possible, seeking help from your instructor when needed. (Refer back to Barroga and Matanguihan (2022) for more help. If your project has not progressed to this stage, you should come back to this activity later.)